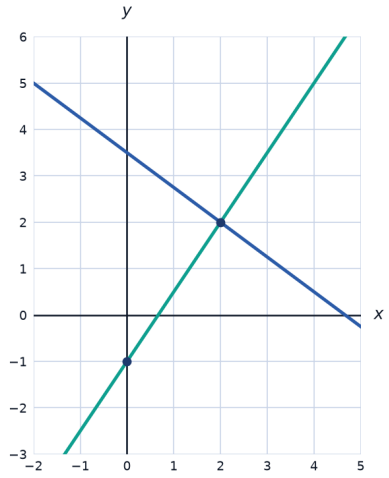
# Simultaneous Equations and Inequalities

## Section 1: Linear-Quadratic Simultaneous Systems

**1** Solve the simultaneous system $y - 2x = 3$ and $x^2 + y^2 = 26$ for the positive solution pair $(x, y)$.




**a.** $(1, 5)$

**b.** $(2, 7)$

**c.** $(3, 9)$

**d.** $(4, 11)$

**2** Find the two points of intersection of the line $2x + y = 7$ and the rectangular hyperbola $xy = 6$.

**a.** $(2, 3)$ and $\left(\frac{3}{2}, 4\right)$

**b.** $(1, 5)$ and $(3, 1)$

**c.** $(2, 3)$ and $(3, 2)$

**d.** $(4, -1)$ and $(1, 6)$

**3** Determine the value of $k$ for which the line $x + y = k$ is tangent to the curve $y = \frac{4}{x}$.

**a.** $k = \pm 4$

**b.** $k = 4$ only

**c.** $k = \pm 2$

**d.** $k = 8$

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## Section 2: Quadratic Inequalities in One Variable

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**4** Find the set of values of $x$ satisfying the quadratic inequality $2x^2 - 5x - 3 \geq 0$.

**a.** $x \leq -\frac{1}{2}$ or $x \geq 3$

**b.** $-\frac{1}{2} \leq x \leq 3$

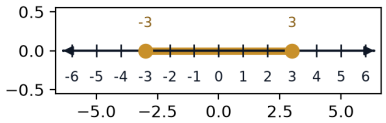
**c.** $x < -1$ or $x > 3$

**d.** $x \geq 3$ only

**5** Solve the strict quadratic inequality $x(x + 4) < 12$.

**a.** $-6 < x < 2$

**b.** $x < -6$ or $x > 2$




**c.** $-2 < x < 6$

**d.** $0 < x < 3$

**6** Find the range of values of $x$ for which the expression $9 - x^2$ is strictly positive.

**a.** $-3 < x < 3$

**b.** $x < -3$ or $x > 3$

**c.** $x < 3$

**d.** $0 < x < 9$

## Section 3: Rational Inequalities and Sign Analysis

**7** Solve the rational inequality $\frac{x-2}{x+3} \leq 0$.

**a.** $-3 < x \leq 2$

**b.** $-3 \leq x \leq 2$

**c.** $x < -3$ or $x \geq 2$

**d.** $x \leq 2$

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**8** Find the complete solution set for the inequality $\frac{2x+1}{x-1} > 1$.

**a.** $x < -2$ or $x > 1$

**b.** $-2 < x < 1$

**c.** $x > 1$ only

**d.** $x > -2$

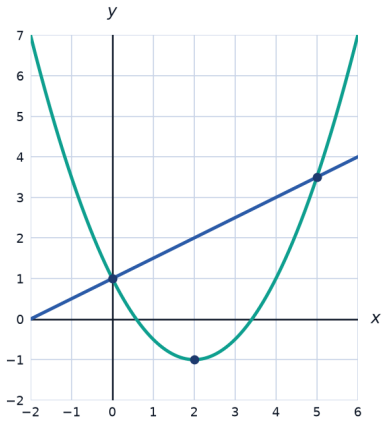
**9** Solve the inequality $\frac{1}{x} < 2$ for all non-zero real $x$.

**a.** $x < 0$ or $x > \frac{1}{2}$

**b.** $0 < x < \frac{1}{2}$

**c.** $x > \frac{1}{2}$ only

**d.** $x < \frac{1}{2}$




## Section 4: Systems with Modulus and Geometric Applications

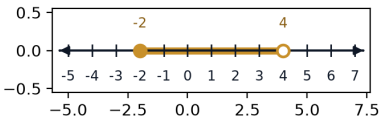
**10** Find the values of $x$ satisfying the simultaneous conditions $|x - 3| \leq 4$ and $x^2 - 4x > 0$.

**a.** $-1 \leq x < 0$ or $4 < x \leq 7$

**b.** $-1 \leq x \leq 7$

**c.** $0 < x < 4$

**d.** $4 < x \leq 7$ only




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**11** The perimeter of a right-angled triangle is $30\ \text{cm}$ and its hypotenuse is $13\ \text{cm}$. Find the lengths of the two legs.

**a.** $5\ \text{cm}$ and $12\ \text{cm}$

**b.** $6\ \text{cm}$ and $11\ \text{cm}$

**c.** $7\ \text{cm}$ and $10\ \text{cm}$

**d.** $8\ \text{cm}$ and $9\ \text{cm}$

**12** Find the range of values of $x$ where the line $y = 3x - 2$ lies entirely above the parabola $y = x^2 - 2x + 4$.

**a.** $2 < x < 3$

**b.** $x < 2$ or $x > 3$

**c.** $-3 < x < -2$

**d.** No such interval exists

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